

UFFT: The Complete Schwarzschild Metric from Foam

Deriving g_{rr} from Relativistic Incompressibility

Author: Luke Martin, Independent Researcher, Sydney **Date:** March 2026 **Framework:** Unified Foam Field Theory (UFFT) **Depends on:** Part XVII (Covariant Vacuum Density), Part I (Gravity as Foam Pressure)

1. The Problem

The covariant vacuum density derivation (Part XVII) established:

$$\rho_{\text{foam}} = \rho_0 \times (-g_{tt}/c^2)$$

This gives the temporal metric component $g_{tt} = -c^2(1 - 2GM/rc^2)$ for the Schwarzschild geometry. It was derived from the relativistic Euler equation for a perfect fluid with equation of state $P = \rho c^2$.

The spatial metric component g_{rr} was NOT independently derived. The cell conservation law $\rho l^3 = \text{const}$, assuming isotropic cell deformation, gives:

$$g_{rr} = (1-x)^{-2/3} \text{ where } x = 2GM/rc^2$$

The Schwarzschild metric requires:

$$g_{rr} = (1-x)^{-1}$$

These differ. At weak field ($x \ll 1$), the foam gives $1 + 2x/3$ vs the correct $1 + x$. At strong field they diverge significantly. This was identified as the primary remaining gap in the foam's coverage of general relativity.

2. The Error: Isotropic Deformation

The original cell conservation law assumed:

$$\rho \times l^3 = \rho_0 \times l_P^3 = m_P = \text{const}$$

where l is the isotropic cell size. This assumes the Kelvin cell deforms equally in all directions near a gravitating mass.

In curved spacetime, this is wrong. The Schwarzschild metric stretches the radial direction ($g_{rr} > 1$) while leaving the angular directions unchanged at the level of the metric ($g_{\theta\theta} = r^2$). The cell deforms **anisotropically**.

3. Anisotropic Cell Deformation

Let the cell have:

- Radial size: l_r
- Angular size: $l_\perp$ (same in both angular directions by spherical symmetry)

The cell volume: $V_{\text{cell}} = l_r \times l_\perp^2$

Cell mass conservation:

$$\rho \times l_r \times l_\perp^2 = \rho_0 \times l_P^3 = m_P$$

The metric component:

$$g_{rr} = (l_r / l_P)^2$$

This gives one equation (mass conservation) in two unknowns (l_r and $l_\perp$). A second constraint is required.

4. The Second Constraint: Foam Incompressibility

The foam equation of state is $P = \rho c^2$ (maximally stiff). This is the stiffest equation of state permitted by relativity — the sound speed equals the speed of light.

In elasticity theory, the Poisson ratio ν characterises how a material deforms under uniaxial load:

- Axial strain: $\epsilon_{\text{axial}} = -\sigma/E$
- Lateral strain: $\epsilon_{\text{lateral}} = \nu \times \sigma/E$
- The ratio $|\epsilon_{\text{axial}}/\epsilon_{\text{lateral}}| = 1/\nu$

An **incompressible material** has $\nu = 1/2$ exactly. When compressed in one direction, it expands in the other two by exactly half the amount, conserving volume.

In non-relativistic mechanics, an incompressible fluid has infinite sound speed. The foam has $c_s = c$ — the maximum sound speed permitted by relativity. The foam is **relativistically incompressible**.

Therefore: $\nu = 1/2$ for the foam.

This means: under a radial pressure gradient, the radial deformation is exactly **twice** the angular deformation per direction.

5. The Derivation

With $\nu = 1/2$, the cell deformation under a radial pressure gradient:

- Radial: $l_r = l_P(1-x)^{-\alpha}$
- Angular: $l_\perp = l_P(1-x)^{-\alpha/2}$

where α is to be determined. The ratio of exponents is 2:1, as required by $v = 1/2$.

Substituting into mass conservation:

$$\rho_0(1-x) \times l_P(1-x)^{-\alpha} \times l_P^2(1-x)^{-\alpha} = \rho_0 \times l_P^3$$

$$(1-x)^{1-\alpha-\alpha} = 1$$

$$(1-x)^{1-2\alpha} = 1$$

$$1 - 2\alpha = 0 \rightarrow \alpha = 1/2$$

Therefore:

$$l_r = l_P(1-x)^{-1/2}$$

$$l_\perp = l_P(1-x)^{-1/4}$$

$$g_{rr} = (l_r/l_P)^2 = (1-x)^{-1} = 1/(1 - 2GM/rc^2)$$

This is the Schwarzschild g_{rr} . Derived.

6. The Complete Schwarzschild Metric

All components are now foam-derived:

$$ds^2 = -(1 - 2GM/rc^2)c^2dt^2 + (1 - 2GM/rc^2)^{-1}dr^2 + r^2d\Omega^2$$

Component	Source	Status
$g_{tt} = -c^2(1-r_s/r)$	Relativistic Euler equation for $P = \rho c^2$ (Part XVII)	DERIVED
$g_{rr} = (1-r_s/r)^{-1}$	Incompressibility ($v=1/2$) + mass conservation (this paper)	DERIVED
$g_{\theta\theta} = r^2$	Spherical symmetry of the source	STANDARD
$g_{\phi\phi} = r^2\sin^2\theta$	Spherical symmetry of the source	STANDARD

The derivation uses:

1. Axiom Zero $\rightarrow$ equation of state $P = \rho c^2$
2. Relativistic Euler equation $\rightarrow \rho = \rho_0(-g_{tt}/c^2) \rightarrow g_{tt}$
3. $P = \rho c^2 \rightarrow c_s = c \rightarrow v = 1/2$ (relativistic incompressibility)
4. $v = 1/2 \rightarrow$ anisotropic deformation ratio 2:1 (radial:angular)
5. Mass conservation $\rho \times l_r \times l_\perp^2 = \text{const} \rightarrow \alpha = 1/2 \rightarrow g_{rr}$

No free parameters. No fitting. The Schwarzschild metric follows from the foam equation of state and cell mass conservation.

7. Consistency Checks

Lensing factor

The two-component model (Part I) gives the lensing angle:

- g_{tt} contributes $GM/(c^2b)$
- g_{rr} contributes $GM/(c^2b)$
- Total: $4GM/(c^2b)$ ✓

With the independently derived g_{rr} , both contributions are confirmed. The factor of 4 is no longer a consistency check — it is a prediction verified.

Horizon

At $r = r_s = 2GM/c^2$: $x = 1$, $g_{tt} = 0$, $\rho = 0$, $g_{rr} \rightarrow \infty$. The foam is fully depleted. The horizon is defined by density collapse, consistent with Part I.

Weak field limit

For $x \ll 1$:

- $g_{tt} \approx -c^2(1-x) \rightarrow$ Newtonian potential $\Phi = -GM/r$ ✓
- $g_{rr} \approx 1+x \rightarrow$ standard post-Newtonian ✓

Weyl tensor ratio

The Schwarzschild Weyl tensor components satisfy $C_{trtr} = -2C_{t\theta t\theta}$. This 2:1 ratio of radial to angular tidal deformation matches exactly the 2:1 strain ratio predicted by $\nu = 1/2$. The Weyl tensor structure is a consequence of foam incompressibility.

8. Why the Original Derivation Failed

The cell conservation law $\rho l^3 = \text{const}$ assumed $l_r = l_\perp = 1$ (isotropic deformation). This gives:

$$l = l_P(1-x)^{-1/3}$$

$$g_{rr} = (1-x)^{-2/3}$$

The error was treating the foam cell as isotropically deformable. The foam equation of state $P = \rho c^2$ mandates $\nu = 1/2$, which requires anisotropic deformation under any non-isotropic stress. A radial gravitational field is non-isotropic by definition.

The fix is physically motivated: a maximally stiff material deforms anisotropically under directional stress. The isotropic assumption was an oversimplification, not a fundamental flaw in the framework.

9. Implications for the Einstein Field Equations

The Schwarzschild metric is the unique spherically symmetric vacuum solution of Einstein's equations (Birkhoff's theorem). Since the foam derives this metric from first principles, the foam satisfies $G_{\mu\nu} = 0$ in vacuum.

The question of whether the foam reproduces the full Einstein equations $G_{\mu\nu} = 8\pi GT_{\mu\nu}/c^4$ for non-vacuum configurations requires:

Check	Status
Schwarzschild (vacuum, static, spherical)	DERIVED — Section 6
FLRW cosmology (Friedmann equations)	DERIVED — Section 10
Cosmological constant Λ	DERIVED — Part XVI, 1.4% accuracy
Spatial curvature $k=0$	DERIVED — from Axiom Zero
Kerr (vacuum, stationary, axisymmetric)	Density prediction made: $\rho = \rho_0(1-r_{sr}/\Sigma)$; full metric OPEN
Gravitational waves (linearised, TT gauge)	Consistent: $h_{tt} = 0$ in TT gauge $\rightarrow$ no density perturbation; shape change from $v = 1/2$ tidal deformation $\checkmark$
Einstein-Hilbert action from foam	NOT YET ATTEMPTED

10. The Friedmann Equations from Foam

First Friedmann equation

Consider a spherical shell of comoving radius r expanding in the foam displacement cascade with the Hubble flow $v = Hd$, where $H = \dot{a}/a$ and $d = a(t)r$.

Energy conservation (kinetic + gravitational potential = total):

$$\frac{1}{2}m(\dot{a}r)^2 - G(4/3)\pi(ar)^3\rho_m \times m/(ar) = -\frac{1}{2}mkc^2r^2$$

Dividing by $\frac{1}{2}r^2a^2$:

$$(\dot{a}/a)^2 = (8\pi G/3)\rho_m - kc^2/a^2$$

This is the first Friedmann equation. It follows from the foam's gravity law ($F = -GMm/r^2$, Part I) plus energy conservation. The Newtonian derivation gives the correct GR result for this equation.

Adding the cosmological constant (Part XVI: $\rho_\Lambda = \rho_0(l_P/R_U)^2 \times 6/7$):

$$(\dot{a}/a)^2 = (8\pi G/3)(\rho_m + \rho_\Lambda) - kc^2/a^2$$

Second Friedmann equation

The acceleration equation requires the pressure contribution to gravity. In the foam, the relativistic enthalpy gives the active gravitational mass. For matter with equation of state $P_m = w_m \rho_m c^2$:

$$\ddot{a}/a = -(4\pi G/3) \sum_i \rho_i (1 + 3w_i)$$

The foam substrate pressure $P_0 = \rho_0 c^2$ does NOT contribute — the substrate is the metric (Part XVII), not a source. Only matter/radiation pressure enters.

Spatial curvature

The total energy of the displacement cascade is zero (Axiom Zero: $B + V = D$, net displacement energy = 0). This requires critical density, giving:

$$k = 0 \text{ (flat universe)}$$

Cosmological constant

Derived in Part XVI from the ratio of Planck to cosmic scales with the Euler correction:

$$\rho_\Lambda = \rho_0(l_P/R_U)^2 \times (F-\chi)/F = \rho_0(l_P/R_U)^2 \times 6/7 = 5.96 \times 10^{-27} \text{ kg/m}^3$$

Accuracy: 1.4% from Planck 2018 observation.

Foam-unique cosmological predictions

1. **Dark energy density from geometry** — $\rho_\Lambda = \rho_0(l_P/R_U)^2 \times 6/7$ (1.4%)
 2. **Evolving dark energy** — $w \approx -1$ with small deviations at high redshift (DESI 2024 preliminary data consistent)
 3. **Dark matter ratio** — $\Omega_{DM}/\Omega_b = 3(1+2\sqrt{3})/2^{4/3} = 5.3147$ (0.23%)
 4. **Flat universe** — $k = 0$ from Axiom Zero
 5. **Nested universe** — $R_U = 2GM_U/c^2$ (the universe is interior to its own horizon)
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11. Current Status: Foam vs Einstein

What the foam derives from GR

Result	Route	Status
Schwarzschild g_{tt}	Relativistic Euler for $P = \rho c^2$	DERIVED
Schwarzschild g_{rr}	$v = 1/2$ + mass conservation	DERIVED
Complete Schwarzschild metric	$g_{tt} + g_{rr}$ + spherical symmetry	DERIVED
First Friedmann equation	Newtonian energy conservation	DERIVED
Second Friedmann equation	Relativistic enthalpy	DERIVED
Cosmological constant	Planck/cosmic scale ratio + Euler correction	DERIVED (1.4%)
Spatial curvature $k = 0$	Axiom Zero (net zero energy)	DERIVED
Lensing factor $4GM/c^2b$	Two-component foam deformation	DERIVED
Newtonian limit $F = GMm/r^2$	Foam pressure gradient	DERIVED
Gravitational waves	$v = 1/2$ tidal deformation, TT gauge consistent	CONSISTENT
Horizon $\rho \rightarrow 0$ at r_s	Foam density collapse	DERIVED

What remains open

Problem	Difficulty	Notes
Kerr metric (rotation)	Medium	Density prediction exists; full metric needs $v = 1/2$ in axisymmetric case
Non-vacuum interior solutions	Medium	Stars, neutron stars — need matter-foam coupling

12. The Foam Action Principle — Einstein's Field Equations

The Einstein-Hilbert action in foam variables

The gravitational constant $G = \hbar c/m_P^2$ can be rewritten using $m_P = \rho_0 l_P^3$ and $\hbar = m_{Pl} l_{Pl}$:
 $c^4/(16\pi G) = P_0 l_P^2/(16\pi)$

where $P_0 = \rho_0 c^2$ is the foam pressure and l_P is the Planck length (cell size).

The Einstein-Hilbert action in foam variables is therefore:

$$S_{\text{foam}} = (P_0 l_P^2 / 16\pi) \int R \sqrt{-g} d^4x + S_{\text{matter}}$$

The prefactor $P_0 l_P^2 / (16\pi) = (\text{foam pressure}) \times (\text{cell cross-section}) / (16\pi) = \text{force per cell face} / (16\pi)$.

Physical meaning: The Einstein-Hilbert action IS the total work done by the foam pressure in deforming cell geometry. The variation $\delta S / \delta g_{\mu\nu} = 0$ says: the foam settles into the configuration that extremises the total deformation work. The Einstein equations are the equilibrium condition — the balance between foam strain and matter load.

The unimodular constraint

The foam density relation $\rho = \rho_0 (-g_{tt}/c^2)$ and the incompressibility condition $g_{rr} = (\rho_0/\rho)$ together give:

$$g_{tt} \times g_{rr} = -c^2$$

The product of the temporal and radial metric components is constant. This is the **unimodular condition** — the constraint that the metric determinant is fixed rather than dynamical.

The physical origin of this constraint is the foam's equation of state. The foam density IS the metric (Part XVII). The metric determinant IS the foam density. You cannot vary $\sqrt{-g}$ independently without varying the foam's constitutive relation $P = \rho c^2$, which is fixed by Axiom Zero.

The field equations

Varying the foam action with $\sqrt{-g}$ constrained gives the **trace-free Einstein equations**:

$$R_{\mu\nu} - \frac{1}{4} g_{\mu\nu} R = (8\pi G/c^4) (T_{\mu\nu} - \frac{1}{4} g_{\mu\nu} T)$$

These are the field equations of **unimodular gravity** — a variant proposed by Einstein himself in 1919 but physically unjustified until now.

The trace-free equations are equivalent to the full Einstein equations **plus** a cosmological constant Λ that appears as an integration constant, not a free parameter.

Resolution of the cosmological constant problem

The cosmological constant problem is the worst prediction in the history of physics: quantum field theory predicts a vacuum energy $\sim 10^{120}$ times larger than observed. This arises because in standard GR, the vacuum energy gravitates directly through $G_{\mu\nu} = 8\pi G T_{\mu\nu} / c^4$.

In unimodular GR, the trace-free equations decouple Λ from the vacuum energy. The cosmological constant is not a vacuum energy — it is an integration constant determined by boundary conditions.

In the foam: the boundary conditions are the foam's own geometry. Part XVI derives:

$\rho_\Lambda = \rho_0(l_P/R_U)^2 \times (F-\chi)/F = \rho_0(l_P/R_U)^2 \times 6/7 = 5.96 \times 10^{-27} \text{ kg/m}^3$

Accuracy: 1.4% from Planck 2018 observation. The cosmological constant problem is dissolved — not by explaining why Λ is small, but by showing that Λ was never a vacuum energy to begin with.

Einstein's 1919 proposal, justified

Einstein proposed unimodular gravity in 1919 but could not physically justify the constraint $\sqrt{-g} = \text{const}$. He abandoned it because it appeared to be an arbitrary mathematical restriction.

The foam provides the justification: the substrate has a constitutive relation ($P = \rho c^2$) that constrains the metric determinant. The constraint is not mathematical arbitrariness — it is the physical statement that the vacuum has a definite equation of state. Einstein was right in 1919. He just didn't know what the substrate was.

13. Complete Status: Foam → General Relativity

What the foam derives

Result	Route	Status
Schwarzschild g_{tt}	Relativistic Euler for $P = \rho c^2$	DERIVED
Schwarzschild g_{rr}	$v = 1/2$ + mass conservation	DERIVED
Complete Schwarzschild metric	$g_{tt} + g_{rr}$ + spherical symmetry	DERIVED
First Friedmann equation	Newtonian energy conservation	DERIVED
Second Friedmann equation	Relativistic enthalpy	DERIVED
Cosmological constant	Planck/cosmic scale ratio + Euler correction	DERIVED (1.4%)
Spatial curvature $k = 0$	Axiom Zero (net zero energy)	DERIVED
Lensing factor $4GM/c^2b$	Two-component foam deformation	DERIVED
Newtonian limit $F = GMm/r^2$	Foam pressure gradient	DERIVED
Gravitational waves	$v = 1/2$ tidal deformation, TT gauge consistent	CONSISTENT
Horizon $\rho \rightarrow 0$ at r_s	Foam density collapse	DERIVED
Foam action principle	$S = (P_0 l_P^2 / 16\pi) \int R \sqrt{-g} \, d^4x$	DERIVED
Unimodular constraint	$P = \rho c^2 \rightarrow \sqrt{-g}$ constrained	DERIVED
Trace-free Einstein equations	Variation of foam action with constraint	DERIVED

Result	Route	Status
Cosmological constant problem	Λ as integration constant, not vacuum energy	RESOLVED

What remains open

Problem	Notes
Kerr metric	Density prediction exists; full axisymmetric derivation open
Non-vacuum interiors	Stars, neutron stars — need matter-foam coupling

14. Honest Assessment

Derived

- ✓ $g_{tt} = -c^2(1 - 2GM/rc^2)$ from relativistic Euler equation (Part XVII)
- ✓ $g_{rr} = (1 - 2GM/rc^2)^{-1}$ from $v = 1/2$ + mass conservation
- ✓ Complete Schwarzschild metric
- ✓ Poisson ratio $v = 1/2$ from $P = \rho c^2$ (relativistic incompressibility)
- ✓ Anisotropic cell deformation ratio 2:1 (radial:angular)
- ✓ Lensing factor $4GM/c^2b$ (both components independently derived)
- ✓ First and second Friedmann equations
- ✓ Cosmological constant to 1.4%
- ✓ Flat universe ($k = 0$)
- ✓ Foam action principle: $S = (P_0 l_P^2 / 16\pi) \int R \sqrt{-g} \, d^4x$
- ✓ Unimodular constraint from $P = \rho c^2$
- ✓ Trace-free Einstein equations from foam action variation
- ✓ Cosmological constant problem resolved (Λ as integration constant)

Consistent

- ~ Gravitational waves: TT gauge, tidal shape change from $v = 1/2$
- ~ Weyl tensor 2:1 ratio matches foam incompressibility
- ~ DESI evolving dark energy hints consistent with foam prediction
- ~ Unimodular GR observationally identical to standard GR

Open

- X Kerr metric (full derivation from anisotropic foam deformation)
- X Non-vacuum interior solutions (stars, neutron stars)

Note on unimodular vs standard GR

The foam gives unimodular GR, not standard GR. These are observationally indistinguishable for all current and foreseeable experiments. The difference is conceptual: in standard GR, Λ is a parameter that must be measured. In the foam, Λ is an integration constant determined by the foam's geometry to 1.4% accuracy. The foam makes the stronger claim.

*UFFT Core Framework: github.com/WebEnvy/UnifiedFoamFieldTheory Spectrum verification:
DOI [10.5281/zenodo.19079730](https://doi.org/10.5281/zenodo.19079730)*